CUSTOMER CHURN ANALYSIS

Telecommunications Customer Churn

Business Analytics & Retention Strategy

From historical churn patterns to a practical management decision and a controlled retention pilot.

7,043	26.54%	4
CUSTOMERS	OBSERVED CHURN	DASHBOARD PAGES

Portfolio report | British English

Completed analysis and proposed future actions are clearly distinguished.

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Reading note: The project is based on fictional sample customer data. The proposed retention pilot has not been implemented, and no causal or realised financial outcome is claimed.

1. Executive Summary

Business challenge

Customer churn can put pressure on recurring income for a subscription-based telecommunications provider. This independent portfolio case study examines **7,043 historical customer records** from the IBM Telco Customer Churn sample dataset. The purpose is to identify where recorded cancellations are concentrated, assess the billing associated with churned customers, and recommend a sensible next step for management.

The dataset contains **1,869 churned customers**, representing **26.54%** of the records analysed. This is an observed proportion in the dataset, not a monthly or annual churn rate; the observation period is not available.

Main findings

Finding	Result	Why it matters
Contract type	42.71% observed churn for month-to-month customers	A clear group to consider for retention testing
Customer tenure	47.44% observed churn at 0-12 months	Early customer relationships deserve attention
Monthly charges	33.98% observed churn for high-charge customers	Charges help define a more focused segment
Billing exposure	86.86% of churn-associated monthly billing relates to month-to-month contracts	Exposure is heavily concentrated
Combined contract and charge segment	50.98% of total churn-associated monthly billing comes from month-to-month, high-charge customers	A major share of exposure lies in one segment
Narrow priority segment	73.49% observed churn among first-year, month-to-month, high-charge customers	A practical starting point for testing a targeted intervention

Important distinction: The **50.98%** figure is the *share of historical billing exposure* from all month-to-month, high-charge churned customers. The **73.49%** figure is the *observed churn proportion* within the narrower first-year, month-to-month, high-charge group. They do not describe the same population or measure.

Python tests found statistically significant associations between churn and contract type, tenure and monthly charges. A logistic regression model achieved a **ROC-AUC of 0.8239** on a held-out test sample, suggesting useful historical ranking performance. The model has not been deployed or validated on a future customer population.

Recommendation to management

Approve the preparation of a limited, controlled proactive-support pilot for currently active customers with month-to-month contracts, 0-12 months' tenure and monthly charges above 80 in the dataset's unspecified currency units.

The intervention would consist of a structured service check-in, help with problems raised by customers and appropriate follow-up. It should be tested against business-as-usual support, ideally using random assignment.

Management should confirm that enough eligible customers are available and agree the costs and success criteria before launch.

The decision to expand should depend on whether the pilot reduces subsequent cancellations **and** delivers sufficient incremental financial value after costs. The historical analysis supports *where to focus*; it does not prove why customers leave or which intervention will work.

Conclusion

The analysis offers a clear way to prioritise retention efforts without committing immediately to a costly, company-wide programme. A controlled test provides the next useful piece of evidence: whether targeted proactive support improves outcomes at a worthwhile cost.

2. Business Context & Objectives

2.1 Business context

Cancellations can affect recurring billing and create a need for additional customer acquisition. A telecommunications provider therefore needs to know which groups are most associated with churn and where its retention resources could be used well.

This report uses a publicly available sample dataset representing a fictional telecommunications business. It is an **independent, simulated consulting engagement**, not work completed for a real client. No genuine client budget, performance target or implementation result has been supplied.

The aim is to turn historical customer information into a management recommendation while being clear about what the data cannot tell us.

2.2 Problem definition

The records show **1,869 churned customers out of 7,043**, or **26.54%**. The practical question is not simply how many customers left. Management needs to know where cancellations are concentrated, how much historical monthly billing is associated with those cancellations, and which customer groups should be considered first for retention testing.

The dataset does not identify the observation period, confirmed cancellation reasons, current customer status, realised revenue loss or intervention costs. For that reason, the report focuses on evidence-based prioritisation rather than an unsupported claim that the business can recover a particular amount of revenue.

2.3 Primary decision question

Which customer segments should management prioritise for retention interventions, and how should it judge whether those interventions reduce subsequent cancellations at an economically worthwhile cost?

2.4 Project objectives

1. **Establish the baseline.** Calculate the overall observed churn proportion and validate customer counts.
2. **Identify priority groups.** Compare churn across contract type, tenure and monthly charge bands, including combinations of these characteristics.
3. **Measure billing exposure.** Calculate the historical monthly charges associated with churned records and assess where those amounts are concentrated.
4. **Assess relationships and prediction.** Test the main statistical associations and evaluate a simple logistic regression model on held-out data.
5. **Support a management decision.** Recommend a defined intervention to test, with responsibilities, costs and outcome measures.

2.5 Analytical hypotheses

Hypothesis	Question	Method
H1: Contract type	Is churn status associated with contract type?	Chi-square test; Cramer's V
H2: Customer tenure	Do churned and non-churned customers have different tenure distributions?	Mann-Whitney U; rank-biserial effect size
H3: Monthly charges	Do the two groups have different monthly charge distributions?	Mann-Whitney U; rank-biserial effect size

These hypotheses examine associations. None assumes that changing a contract, increasing tenure or reducing a bill would, by itself, prevent a cancellation.

2.6 Scope and proposed measures of success

The completed work covers data preparation, SQL exploration, historical billing segmentation, statistical tests, logistic regression and a four-page Power BI report. Live customer targeting, intervention delivery, customer contact outcomes and financial benefits are **not** part of the completed work.

For a future pilot, management should agree three main outcomes before launch: the difference in subsequent cancellation rates between assigned groups, the incremental commercial contribution created by improved retention, and the cost of achieving that result. Targets and budgets should come from current operational information, not be invented from the historical dataset.

3. Data & Methodology

3.1 Source and dataset

The project used the **IBM Telco Customer Churn** sample dataset. It contains **7,043 customer records and 21 original columns** covering customer details, subscriptions, services, tenure, monthly charges, total charges and a binary churn label.

The most important predictors for this case were contract type (**Month-to-month**, **One year**, **Two year**), tenure in months and monthly charges. The target field, **Churn**, records whether the customer was classified as having left. The data does not specify a currency, so this report does not attach a currency symbol to charge values.

The work followed four stages: **Excel Power Query -> MySQL Workbench -> Python in Google Colab -> Power BI**.

3.2 Excel Power Query: clean and prepare

The first stage checked the raw dataset for missing values, duplicates, numerical errors and inconsistent categories. The aim was to avoid drawing business conclusions from incorrectly prepared records.

Missing values. The original **TotalCharges** field contained **11 missing values**. All 11 belonged to records with **zero months of tenure**. A new column, **TotalCharges_Clean**, was created with these missing values set to zero. This was a documented, practical assumption for preparing the data; it was not confirmation of what those customers had actually paid. No missing values remained in the new cleaned field.

Duplicate checks. No duplicate customer IDs were found. The prepared dataset contained **7,043 unique customers**.

Type and range checks. Tenure, monthly charges and cleaned total charges were checked as numeric fields. The validated ranges were:

Field	Minimum	Maximum
Tenure	0	72
MonthlyCharges	18.25	118.75
TotalCharges_Clean	0	8,684.80

Contract, churn, internet service and payment method categories were reviewed for consistent labels. Once the checks were complete, the cleaned data was exported as a UTF-8 CSV for use in SQL.

3.3 MySQL Workbench: explore and validate

After importing the CSV into MySQL Workbench, I checked the row count, distinct customer IDs, variable types, categorical frequencies and numerical ranges against the Power Query results. The import contained **7,043 records**, and the ID check confirmed **7,043 unique customers**.

Numbered and commented SQL queries were used to calculate total and segment-level customer counts, churn proportions and historical monthly billing associated with churned customers.

The questions followed a practical order. First, establish overall churn. Next, compare churn by contract type, tenure and monthly charge band. Then examine combinations of those variables, such as contract by charge band and contract by tenure. Finally, compare the shares of churn-associated monthly billing across these segments.

The billing measure sums **MonthlyCharges** for records marked **Churn = Yes**. It is a **historical monthly billing exposure measure**, not a verified loss in a particular month, an annual forecast or an estimate of recoverable revenue.

3.4 Python: test associations

Google Colab and Python were used to assess whether the patterns observed in SQL were supported by statistical evidence. A chi-square test was applied to contract type and churn. Mann-Whitney U tests compared tenure and monthly charges between churned and non-churned records.

Effect sizes were reported alongside the p-values. For contract type, **Cramer's V = 0.4101**. For tenure and monthly charges, the rank-biserial estimates were **r = -0.4797** and **r = +0.2416**, respectively. All three tests reported **p < 0.001**.

These are findings about statistical associations. They do not establish the causes of cancellation. The tests are discussed in more detail in Section 5.

3.5 Logistic regression: model development

A regularised logistic regression model was fitted using three predictors: contract type, tenure and monthly charges. Contract type was represented by indicator variables with the two-year contract as the comparison category. Numerical predictors were standardised within the modelling process.

An **80/20 stratified split** gave **5,634 training records** and **1,409 held-out test records**. Training data was used for model development and five-fold cross-validation. The default classification threshold of **0.50** was compared with a provisional threshold of **0.40** selected during training-stage evaluation. Both were assessed on the held-out test sample.

At threshold 0.40, accuracy was **75.80%**, precision **53.72%**, recall **63.64%** and F1-score **58.26%**. The model's threshold-independent ranking measures were **ROC-AUC 0.8239** and **Average Precision 0.5978**. These ranking measures are calculated across decision thresholds, rather than being results specific to 0.40.

The model was not deployed. Probability calibration, testing on later-period records and commercial effects remain unverified.

3.6 Power BI: bring the findings together

Power BI was used to build an interactive, four-page report. DAX measures calculated observed churn, historical billing exposure and shares of exposure. The Python test and model results were added as supporting data tables.

Report page	Purpose
1. Overview	Overall customer churn and differences by contract, tenure and charge band
2. Billing	Historical billing exposure, combined segments and the priority customer group
3. Insights	Statistical evidence, adjusted odds ratios and proposed management actions
4. Diagnostics	Model metrics, threshold comparison, confusion matrix and ranking performance

Customer totals and key group figures were cross-checked between the prepared dataset, SQL outputs and Power BI. The Python test results were compared with the displayed diagnostics. This reduced the risk of inconsistent figures appearing in the final report.

3.7 Methodological boundaries

The dataset is historical and the observation window is unknown. It cannot support monthly or yearly churn claims, a full revenue-loss estimate or conclusions about why individual customers cancelled. The predictive model was evaluated on a historical hold-out sample, not a future customer population. These limits affect the strength of the recommendation and are addressed again in Section 8.

4. Key Business Findings

4.1 Overall churn: a clear reason to prioritise

Of **7,043** customers, **1,869** were labelled churned and **5,174** were labelled non-churned. The observed churn proportion was **26.54%**.

Status	Customers	Share
Churned	1,869	26.54%
Non-churned	5,174	73.46%
Total	**7,043**	**100.00%**

What it means. More than one in four records were classified as churned. This warrants closer examination, but it is not an annual or monthly churn rate because the observation period is unavailable.

Management action. Treat 26.54% as the descriptive baseline for this dataset. Focus next on identifying priority segments instead of launching an untargeted retention programme.

4.2 Contract type: monthly customers stand out

Contract	Customers	Churned	Observed churn
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

Month-to-month customers had **42.71% observed churn**, compared with **11.27%** for one-year contracts and **2.83%** for two-year contracts. They also accounted for **1,655 of the 1,869 recorded cancellations**.

What it means. Contract type is strongly associated with the recorded churn outcome. Customers selecting shorter contracts may also differ from longer-term subscribers in other ways; the contract itself is not proved to cause cancellation.

Management action. Prioritise month-to-month customers for retention testing. Longer-term contract incentives are a possible *separate* experiment, not a proven solution or an immediate blanket recommendation.

4.3 Tenure: the first year matters

Tenure group	Customers	Churned	Observed churn
0-12 months	2,186	1,037	47.44%
13-36 months	1,856	474	25.54%
37-72 months	3,001	358	11.93%

Customers in their first **0-12 months** recorded **47.44% churn**, compared with **11.93%** for customers with 37-72 months of tenure. The first-year group accounted for **1,037 recorded cancellations**.

What it means. Shorter-tenure customers are an important group for further attention. The data does not establish whether onboarding, service quality, pricing or another factor explains the difference.

Management action. Include eligible first-year customers in a controlled proactive-support pilot, with feedback recorded to understand what problems they report.

4.4 Monthly charges: higher-charge groups show more churn

Monthly charge band	Customers	Churned	Observed churn
Low: below 40	1,837	213	11.59%
Medium: 40-80	2,540	750	29.53%
High: above 80	2,666	906	33.98%

The high-charge group recorded **33.98% churn**, compared with **11.59%** in the low-charge group.

What it means. Monthly charges are useful for prioritisation, particularly because higher-charge customers contribute more billing per customer. But the analysis does not show that price was the reason for leaving: service packages, expectations and contract arrangements may matter.

Management action. Include the charge band in pilot eligibility rules. Do not introduce broad discounts before customer feedback and financial analysis justify a pricing test.

4.5 Billing exposure: a highly concentrated pattern

The sum of **MonthlyCharges** for all records marked churned was **139,130.85**. Month-to-month contracts accounted for **120,847.10**, or **86.86%** of the total.

The combined **month-to-month, high-charge** segment accounted for **70,928.35**, equivalent to **50.98%** of all historical churn-associated monthly billing.

Segment	Historical churn-associated monthly billing	Share
Month-to-month and high charges	70,928.35	50.98%
All remaining segments	68,202.50	49.02%
Total	**139,130.85**	**100.00%**

What it means. A little over half of the historical monthly billing linked to churned records is concentrated in this combined segment. This is a useful signal for prioritisation, not confirmed revenue lost or a forecast of savings from a retention programme.

Management action. Start with month-to-month, high-charge customers and use tenure to identify a smaller group suitable for targeted testing.

4.6 Narrow priority group: the clearest pilot starting point

Combining the three characteristics identified **464 historical records** for customers with month-to-month contracts, **0-12 months' tenure** and monthly charges **above 80**.

Priority group measure	Result
Historical records	464
Recorded as churned	341
Recorded as non-churned	123
Observed churn proportion	**73.49%**

What it means. The group's **73.49%** observed churn proportion is substantially above the **26.54%** overall figure. The 123 historical non-churned records are **not** a verified list of customers who are active today.

Management action. Use the same transparent eligibility rules to identify currently active customers in the provider's live CRM system. If the available population is too small for a useful trial, consider a broader group such as first-year, month-to-month customers, while analysing the original priority group separately.

4.7 Bringing the findings together

Three broad patterns point in the same direction: recorded churn is higher among month-to-month customers, shorter-tenure customers and higher-charge customers. Billing exposure is particularly concentrated in the month-to-month, high-charge combination. Within that population, first-year customers have an especially high historical churn proportion.

This provides a strong basis for choosing *where* to test a retention approach. It does not yet tell us *which approach works*. The next section examines the statistical evidence and predictive model behind this recommendation.

5. Statistical & Predictive Analysis

5.1 Why the additional analysis was needed

SQL showed which groups had higher churn. Statistical tests helped assess the strength of the observed associations, while logistic regression looked at contract type, tenure and monthly charges together. The predictive evaluation then tested how well those variables distinguished historical churned from non-churned records.

These methods improve the evidence for customer prioritisation. They do **not** show that any particular support intervention will prevent cancellation.

5.2 H1: Contract type and churn

A **chi-square test** examined whether contract type and churn status were related. It produced **p < 0.001**, while **Cramer's V = 0.4101** indicated a meaningful association.

The test supports the clear descriptive difference: 42.71% observed churn for month-to-month contracts compared with 2.83% for two-year contracts. It does not demonstrate that changing an individual's contract will cause them to stay.

5.3 H2: Tenure and churn

A **Mann-Whitney U test** compared tenure between churned and non-churned customers. The reported result was **p < 0.001**, with a **rank-biserial correlation of -0.4797**.

The negative sign reflects the tendency for churned records to have shorter tenure. It is an effect-size estimate, **not** the Mann-Whitney U test statistic itself. The evidence supports prioritising early-tenure customers, but it does not identify the reason they cancelled.

5.4 H3: Monthly charges and churn

A second **Mann-Whitney U test** compared monthly charges. It reported **p < 0.001**, with a **rank-biserial correlation of +0.2416**.

The direction is consistent with higher charges being more common among churned records. As with tenure, the correlation is an effect-size estimate, not the U statistic. The result supports including charges in prioritisation; it does not prove price caused cancellation.

Hypothesis	Test	Effect size	P-value
H1: Contract type	Chi-square	Cramer's V = 0.4101	< 0.001
H2: Customer tenure	Mann-Whitney U	Rank-biserial r = -0.4797	< 0.001
H3: Monthly charges	Mann-Whitney U	Rank-biserial r = +0.2416	< 0.001

The tests' statistical significance should not be confused with proof of causation or proof that a retention intervention will succeed.

5.5 Logistic regression: the predictors considered together

A regularised logistic regression model included contract type, tenure and monthly charges. The adjusted odds ratios describe the association with recorded churn **after accounting for the other included predictors**.

Characteristic	Adjusted odds ratio	Plain-English interpretation
Month-to-month vs two year	7.10	Around 7.10 times the estimated odds of churn
One year vs two year	2.36	Around 2.36 times the estimated odds
12 more months of tenure	0.66	Around 34% lower estimated odds
10 more units of monthly charges	1.34	Around 34% higher estimated odds

These are **odds**, not direct changes in churn probability. The estimates are model-based associations, not causal effects. The model used regularisation; separate coefficient-significance claims have not been established.

5.6 Performance at two classification thresholds

The model was evaluated on **1,409 held-out records**. Four threshold-dependent metrics were compared:

Classification measure	Threshold 0.50	Threshold 0.40
Accuracy	77.71%	75.80%
Precision	59.80%	53.72%
Recall	48.93%	63.64%
F1-score	53.82%	58.26%

Accuracy is the overall proportion classified correctly. **Precision** measures how many predicted churners were actually churned. **Recall** measures how many of the historical churners were found. **F1-score** balances precision and recall.

Lowering the classification threshold from 0.50 to 0.40 increased recall from **48.93% to 63.64%** but reduced precision from **59.80% to 53.72%**. This is a business trade-off: it may help identify more potential cancellations but also means contacting more customers who would not otherwise be flagged correctly.

5.7 Confusion matrix: what those trade-offs look like

At the provisional **0.40** threshold, the results were:

Actual / predicted	Predicted no	Predicted yes	Total
Actual no	830	205	1,035
Actual yes	136	238	374
Total	**966**	**443**	**1,409**

The model correctly classified **830 non-churned** and **238 churned** records. It incorrectly flagged **205 non-churned** records and missed **136 churned** records. Compared with threshold 0.50, threshold 0.40 detected **55 additional historical churners** but created **82 additional false positives**.

The correct operational threshold would depend on intervention costs and the financial value of preventing cancellations. The 0.40 threshold is a reasonable candidate for evaluation, **not** an established commercial optimum.

5.8 Ranking performance across thresholds

The held-out sample produced **ROC-AUC = 0.8239** and **Average Precision = 0.5978**. The churn prevalence reference was **0.2654**.

ROC-AUC measures how well the model ranks churned records above non-churned records across possible thresholds. It does **not** mean the model is correct 82.39% of the time. Its chance-level reference is 0.5.

Average Precision summarises precision and recall across thresholds. The test-set churn prevalence of approximately 0.2654 is a useful reference for this measure. These two ranking scores are **not specific to the classification threshold of 0.40**.

Taken together, the results suggest useful historical ranking performance. They do not establish that predicted probabilities are sufficiently calibrated for use in live decisions, or that performance will hold in a future customer population.

5.9 What the evidence adds to the management decision

The statistical tests and logistic regression support the main descriptive patterns. Contract type, tenure and monthly charges contain useful information for choosing which customer groups to prioritise.

They do not prove the reason for cancellation and do not show that proactive support will work. The proposed pilot should therefore use simple, clear segment rules initially. Live predictive scoring should only be considered after further validation and an assessment of operational value.

6. Management Recommendations

6.1 Primary recommendation: a controlled, targeted pilot

Prepare a limited proactive-support pilot for currently active first-year, month-to-month, high-charge customers rather than launch a company-wide retention programme.

This is a proportionate response to the findings. The target segment is supported by the historical churn analysis, while the pilot will test an intervention whose effectiveness is still unknown.

The immediate management decision is to **authorise pilot preparation**, subject to an eligibility review, a feasible sample, an agreed intervention, a defined budget and a suitable evaluation plan. Authorising preparation is not the same as approving an unconditional launch.

6.2 Confirm the current eligible population

Lead: CRM and Data Analytics. Query the live customer management system for active customers with a month-to-month contract, 0-12 months' tenure and monthly charges above 80 in the corresponding business units.

Check customer status, valid contact details, appropriate service records and permissible use of customer data. The historical dataset's **123 non-churned** records are a reason to review eligibility, **not** a confirmed list of current customers to contact.

If numbers are too small for a reliable comparison, consider broadening eligibility to first-year month-to-month customers, while preserving separate reporting for the original target segment.

Output: A verified eligible population, initial operational feasibility assessment and proposed sample-size calculation.

6.3 Test proactive support, not blanket discounts

Lead: Customer Service and Customer Retention. Offer the selected intervention group a structured service check-in through a suitable, permitted channel. Ask about their experience, help with issues they raise and record necessary follow-up.

Staff may address billing queries, technical concerns or service questions **if customers raise them**. These are possible areas of support, not proven causes of churn.

A service-focused pilot is a practical starting point because it can be tested without automatically changing prices or contract terms. It is not guaranteed to work and must be judged on actual cancellation outcomes.

Output: A defined contact script or approach, issue-recording process and clear service responsibilities.

6.4 Compare outcomes fairly

Lead: Data Analytics, supported by CRM. Where practical, randomly assign eligible customers to an intervention group and a business-as-usual comparison group. The latter should retain full access to normal support.

Set the main outcome, observation window and analysis plan before the intervention begins. Count customers in the group to which they were originally assigned when evaluating cancellations, including people whom the team could not reach. This prevents contact success from distorting the result.

Output: A controlled pilot design and an agreed decision framework.

6.5 Judge commercial value as well as cancellations

Lead: Finance and Data Analytics. Evaluate whether the intervention lowers subsequent cancellations compared with normal service. Then estimate whether any additional retention creates enough **incremental contribution**, after intervention costs, to justify expansion.

Include staff time, customer contact, additional support, incentives if any, and the costs of serving customers who remain. Do not treat historical billing exposure as profit, confirmed revenue loss or a forecast of recoverable value.

Output: A balanced commercial assessment with the evidence needed for a scale, revise or stop decision.

6.6 Other opportunities: keep them separate

Voluntary longer-term contracts. Longer contracts are associated with lower historical churn. Management could test suitable voluntary offers, but should not assume contract switching causes lower churn.

Early customer experience. Shorter tenure is associated with more churn. Management could review onboarding and first-year support, but should identify specific service issues before redesigning the entire journey.

Pricing tests. Higher charges are associated with churn, but the data does not show that price caused cancellations. Broad discounts are not recommended. Pricing offers would require their own customer evidence and financial test.

These are possible later experiments, not additional actions claimed to have been implemented.

6.7 Proposed ownership

Proposed team	Responsibility
Head of Customer Retention	Sponsor the pilot and approve its scope
CRM / Data	Confirm eligibility and prepare assignments
Customer Service	Deliver structured check-ins and record outcomes
Data Analytics	Design evaluation and assess results
Finance	Assess cost and incremental commercial value
Legal / Data Protection	Review appropriate data use and customer contact

These responsibilities are proposed for a simulated engagement, not confirmed positions within a real client company.

6.8 Decision requested

Management should **approve feasibility and pilot design work** for the priority segment, with permission to return for launch approval once the population, sample size, intervention cost and success criteria have been checked.

If a pilot is launched, management should then decide to **expand** only with credible evidence of worthwhile results, **revise** where there is a clear correctable issue, or **stop** if the approach is ineffective or uneconomic.

7. Pilot Implementation Plan & Success Measures

7.1 Purpose and proposed population

The pilot should answer a practical question: **Does a structured proactive-support check-in reduce subsequent cancellations enough to justify its cost?**

The starting population is currently active customers with month-to-month contracts, 0-12 months' tenure and monthly charges above 80. The historical **464 records** in this segment, including **123 labelled non-churned**, cannot establish the number eligible today. Eligibility must be checked in live CRM records.

Before launch, the data team should calculate whether the available sample can detect a commercially meaningful difference. The historical observed churn proportion is not a valid substitute for a time-matched prospective cancellation baseline.

7.2 Design: intervention and comparison groups

Where feasible, assign eligible customers at random to two groups:

Group	Experience
Intervention	Structured proactive service check-in plus existing standard support
Comparison	Existing standard support without the additional proactive check-in

Normal customer service must remain available to both groups. Record all assigned customers, contact attempts, completed check-ins, subsequent cancellations and relevant costs.

Use an **intention-to-treat** comparison for the main result: evaluate customers according to the group they were assigned to, even if a check-in did not take place. Contact completion is a separate measure of operational delivery.

If random assignment cannot be used, document the alternative comparison method and the additional uncertainty it creates.

7.3 Proposed timetable

The following schedule is **illustrative**. It is not an agreed client timetable.

Phase	Indicative timing	Main work
Feasibility	Weeks 1-2	Validate current eligible population, data quality and sample size
Preparation	Weeks 3-4	Agree approach, costs, assignment and evaluation plan
Delivery and observation	From week 5	Contact assigned customers and monitor outcomes
Evaluation	After observation	Compare cancellations, delivery and financial outcomes

Phase	Indicative timing	Main work
Management decision	After evaluation	Expand, revise or stop based on agreed criteria

A **90-day observation window** could be considered, but only if it fits the provider's billing and cancellation patterns and is long enough to produce useful evidence. The schedule and follow-up period must be settled before launch.

7.4 Five success KPIs

KPI 1 - Cancellation rate by assigned group. The percentage of customers assigned to each group who cancel within the agreed observation period:

Cancellation rate (%) = cancellations in group / customers assigned to group x 100

KPI 2 - Incremental retention effect. The difference in cancellation rates between the comparison and intervention groups, expressed in **percentage points**:

Cancellation-rate difference = comparison rate - intervention rate

For illustration only, a comparison rate of 20% and intervention rate of 16% would give a **4-percentage-point** difference. No such result has been observed in this project. Report uncertainty around the effect and follow the pre-agreed evaluation plan.

KPI 3 - Completed check-in rate. The share of all intervention-assigned customers who complete the planned check-in:

Completed check-in rate (%) = completed check-ins / customers assigned to intervention x 100

Track contact attempts and refusals separately. Do **not** exclude unreachable customers from the main cancellation comparison.

KPI 4 - Cost per assigned intervention customer. The average additional cost of running the intervention:

Cost per assigned customer = total additional intervention cost / intervention-group customers assigned

Include staff time, contact costs, additional support and any incentives. Finance should agree the cost categories before launch.

KPI 5 - Net incremental commercial value. Estimate the additional contribution created by the intervention, compared with normal service, less the intervention's additional costs:

Net incremental value = incremental contribution from improved retention - additional intervention costs

This requires an estimate of additional customers retained, a suitable contribution margin and an agreed time horizon. It cannot be calculated reliably using historical monthly billing alone.

7.5 Monitoring and evaluation

During delivery, report assigned customer numbers, contact attempts, completed check-ins, issues raised and resolved, and costs. At the agreed evaluation date, compare subsequent cancellations between assigned groups, estimate the size and uncertainty of any difference, and assess its commercial value.

Avoid concluding that a pilot has succeeded based only on happy customer feedback or an early fluctuation in cancellation rates. The primary result must reflect the comparison agreed before launch.

7.6 Risks and safeguards

Risk	Practical response
Too few eligible customers	Run a sample-size check and consider a pre-agreed broader segment
Low response to contacts	Record delivery clearly, but retain original group assignments in evaluation
Higher support costs	Set a capped budget and track full incremental costs
Wrong intervention for the cause of churn	Record customer concerns and avoid claiming known causes
Privacy or inappropriate contact	Obtain legal / data-protection review before outreach
Unvalidated model for live scoring	Use transparent segment rules initially; validate the model separately

7.7 Expand, revise or stop

Expand only if the pilot produces credible evidence of fewer cancellations and sufficient net value, with acceptable operational and customer outcomes.

Revise if the result is uncertain or the intervention had a fixable delivery problem, but only where there is a reason to expect a better outcome from another test.

Stop if the programme shows no meaningful benefit, creates unacceptable harm or costs more than it is expected to generate.

No numerical success target or financial return is asserted in this case study. The client would need to agree those criteria using real operating information.

7.8 Expected deliverables after a future pilot

The proposed future client deliverables are: a verified participant summary; intervention delivery measures; a fair comparison of cancellation rates; uncertainty around the estimated effect; total intervention costs and incremental contribution; and a written recommendation to expand, revise or stop.

These are proposed future outputs, not work claimed to have been completed.

8. Limitations, Next Steps & Final Conclusion

8.1 What the evidence supports

The completed analysis identifies clear historical differences in churn by contract type, tenure and monthly charges. It also shows where monthly billing associated with churned records is concentrated. These findings support a decision about which customer group should be considered first for retention testing.

They do not prove the causes of churn, show that a particular support intervention works, or establish the financial return from a live programme.

8.2 Data limitations

Unknown observation period. The 26.54% observed churn proportion cannot be described as a monthly, quarterly or annual rate.

No verified cancellation reasons. The dataset cannot tell us whether churn was driven by pricing, service quality, onboarding or another factor.

Current status is unknown. Historical non-churned records, including the 123 in the priority segment, are not verified active accounts today.

Incomplete commercial information. Contribution margin, costs of retention, customer acquisition costs and realised revenue loss are not available. Historical churn-associated monthly billing must not be treated as lost profit or recoverable revenue.

Documented cleaning assumption. Eleven missing **TotalCharges** values for zero-tenure records were set to zero in a cleaned field. Original billing records were unavailable to confirm them.

8.3 Statistical and model limitations

The tests establish associations, not causation. Customers on different contracts or at different tenure stages may differ in other important ways. The regularised logistic regression used only three predictors; complaints, usage changes and service problems were not included.

The model was tested on a historical hold-out sample, **not** a future period. It made false positive and false negative classifications, and probability calibration and operational suitability were not established. Live use would require more recent validation and a careful review of costs and error trade-offs.

These limitations are reasons to design a controlled test properly; they are not reasons to abandon a sensible, low-commitment next step.

8.4 Immediate next steps

1. **Confirm the current target population** in CRM using active accounts and the proposed segment criteria.
2. **Assess feasibility**, including eligible numbers, required sample size, staff capacity, budget and legal or data-protection requirements.
3. **Approve the pilot design** only after agreeing the intervention, comparison method, observation period and success criteria.
4. **Measure outcomes** using cancellation differences, delivery rates and incremental contribution after costs.
5. **Make an evidence-based decision** to expand, revise or stop the programme.

Later analytical work could add verified cancellation reasons, complaint history, service usage, customer value and more recent customer records. A revised predictive model could be compared against simple segment rules, but that further modelling is not a prerequisite for assessing pilot feasibility.

8.5 Final management recommendation

This analysis identified where historical churn is concentrated and which customer groups should be considered first for retention efforts. The findings support prioritising first-year, month-to-month customers with high monthly charges.

Management should approve preparation of a controlled proactive-support pilot, not an immediate company-wide rollout. The CRM and data teams should first confirm current eligibility and whether a meaningful evaluation is feasible. If those checks are satisfactory, management should agree the intervention, comparison groups, costs and success criteria before launch.

A wider programme should be considered only if the pilot produces credible evidence of improved customer retention **and** enough incremental value to justify its cost.

This is the central business outcome of the case study: **a clear, proportionate decision informed by the analysis, with a practical way to test whether it works.**

Sources, Definitions & Project Status

Dataset and attribution

Dataset: IBM Telco Customer Churn, publicly available sample customer-churn data representing a fictional telecommunications provider. Original dataset: 7,043 records and 21 columns. No business sponsor or real consulting client is claimed.

Prepared analysis: Excel Power Query, MySQL Workbench, Python/Google Colab and Microsoft Power BI. Figures and analytical results are the completed outputs of the user's portfolio project, as documented across its SQL, Python and dashboard stages.

Important definitions

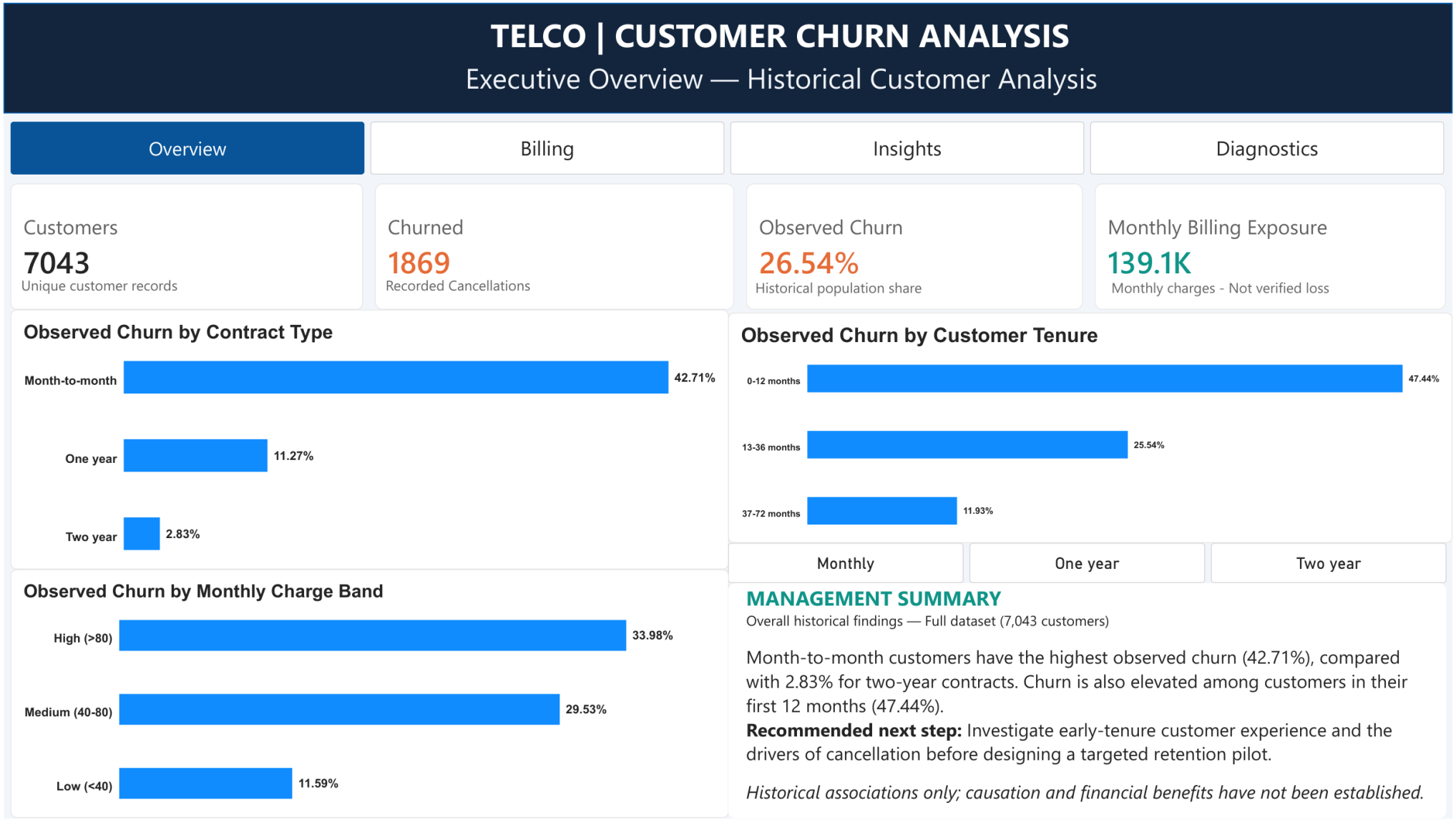
- **Observed churn proportion:** Customers marked **Churn = Yes** divided by all records in the relevant group. No observation period is given.
- **Historical churn-associated monthly billing:** Sum of **MonthlyCharges** for customers marked churned. It is not confirmed revenue lost, a monthly churn rate or a prediction of savings.
- **High monthly charge band: MonthlyCharges > 80**; medium 40-80; low below 40, based on the project's defined bands.
- **Billing segment share:** A selected churn-associated billing total divided by the full churn-associated billing total (139,130.85).
- **Ranking metrics:** ROC-AUC and Average Precision summarise ranking across decision thresholds and are not threshold-0.40-only results.
- **Proposed intervention:** A plan for future client testing. No implementation, trial result or realised financial benefit is claimed.

Dashboard pages referenced

Appendix A reproduces the final Power BI screenshots supplied with this project: (1) Executive Overview, (2) Billing Exposure, (3) Management Insights and (4) Model Diagnostics. The report interprets and references those analyses but does not claim that the screenshots represent live customer operations.

Page 1 | Executive Overview

Observed churn, customer counts, contract, tenure and charge-band comparisons



Page 2 | Billing Exposure

Historical churn-associated monthly billing and priority customer segments

TELCO | CUSTOMER CHURN ANALYSIS

Billing Exposure - Historical Monthly Charges

Overview | **Billing** | Insights | Diagnostics

Churn-Associated Monthly Billing
139,130.85

Month-to-month Share
86.86%

Largest segment share
50.98%

Billing Exposure by Customer Segment

Contract	Low	Medium	High	Total
Month-to-month	3.56%	32.32%	50.98%	**86.86%**
One year	0.20%	2.04%	7.91%	**10.15%**
Two year	0.08%	0.45%	2.46%	**2.99%**
Total	**3.84%**	**34.81%**	**61.34%**	**100.00%**

Share of Churn-Associated Monthly Billing by Contract and Charge Band



Priority Customer Segment

73.49%

Monthly . High charge . 0-12 months

123 historical non-churned records for eligibility review

MANAGEMENT TAKEAWAY

Month-to-month contracts account for **86.86%** of historical churn-associated monthly billing. The month-to-month, high-charge segment represents **50.98%** of overall exposure.

Recommendation: Investigate early-tenure, high-charge customers and verify the eligibility of the **123 historical non-churned records** before considering a targeted retention pilot.

Historical billing exposure does not represent confirmed revenue loss. No retention intervention has been implemented.

TELCO | CUSTOMER CHURN ANALYSIS

Management Insights - Statistical Evidence & Recommendations

Overview | Billing | **Insights** | Diagnostics

Highest Contract Risk	Highest Tenure Risk	Highest Charge Risk
42.71%	47.44%	33.98%

Statistical Evidence - Hypothesis Testing

H1 — Contract Type — $p < 0.001$
Chi-square test . Cramér's V = 0.4101

H2 — Customer Tenure — $p < 0.001$
Mann–Whitney U test . r = -0.4797

H3 — Monthly Charges — $p < 0.001$
Mann-Whitney U test . r = +0.2416

Adjusted Odds Ratios

Adjusted odds ratios indicate associations, not changes in churn probability or causation.

Month-to-month vs Two year 609.67% higher estimated odds	7.10 x
One year vs Two year 135.73% higher estimated odds	2.36x
Tenure (+12 months) 34.21% lower estimated odds	0.66x
Monthly charges (+10 units) 33.69% higher estimated odds	1.34x

PROPOSED MANAGEMENT ACTIONS

01 Investigate
Validate the currently active target population

02 PILOT
Run proactive service check-ins against a comparison group.

03 Decide
Assess churn reduction and costs before scaling

Analytical limitation: *No intervention has been implemented; effectiveness and commercial benefits remain unproven.*

Page 4 | Model Diagnostics

Logistic regression metrics, confusion matrix and ranking performance

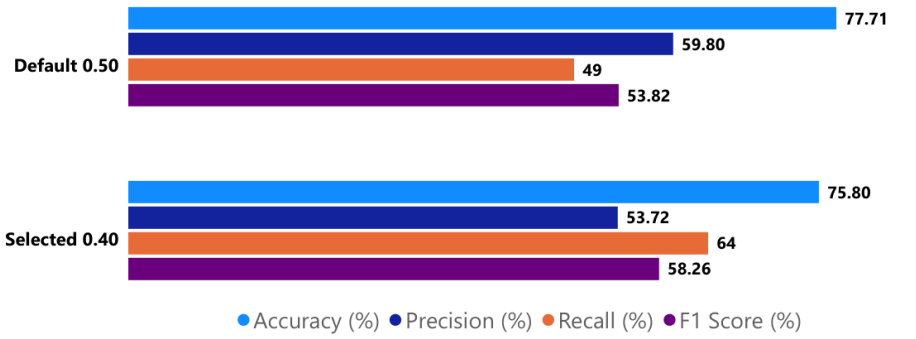
TELCO | CUSTOMER CHURN ANALYSIS

Model Diagnostics - Predictive Performance & Validation

Overview | Billing | Insights | Diagnostics

Accuracy	Precision	Recall	F1 Score
75.80%	53.72%	63.64%	58.26%

Model Performance — Threshold Comparison



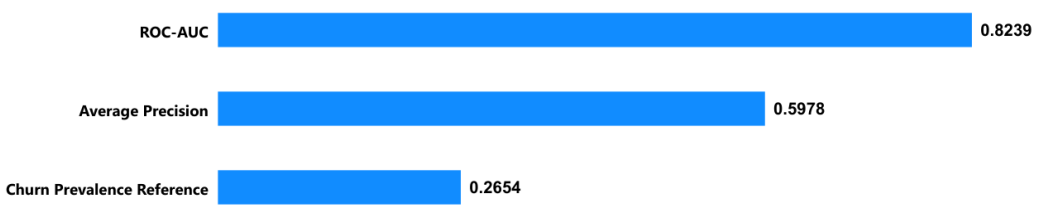
Confusion Matrix — Logistic Regression (Threshold 0.40)

Actual vs Predicted Churn | Held-out Test Sample: 1,409 Customers

Actual Churn	No	Yes	Total
No	830	205	**1035**
Yes	136	238	**374**
Total	**966**	**443**	**1409**

Logistic Regression — Ranking Performance

Average Precision is compared with the churn prevalence reference (0.2654). ROC-AUC has a different chance-level reference of 0.5.



MANAGEMENT INTERPRETATION

At threshold **0.40**, the model identifies **55 more historical churners** than at 0.50, but generates **82 additional false positives**.

Recommendation: Evaluate the costs of missed churners versus unnecessary interventions before selecting a threshold for operational use.

MODEL LIMITATIONS

The model was evaluated on historical held-out data using contract type, tenure and monthly charges. Future-period performance, probability calibration and commercial impact remain unvalidated.

The model has not been deployed, and no retention intervention or financial benefit has been demonstrated.